

Oswald Gyabaah

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 6 years shipping production JavaScript/TypeScript across React/Next.js, Node.js, and AWS serverless (Lambda, EventBridge, DynamoDB). Founded MyHealthCop on AWS EKS + Lambda and led the GTBank Basis-to-Finacle migration cutover across 33 applications in two weeks. Strong in DynamoDB single-table modeling, NoSQL data design across MongoDB and Firestore, and Terraform IaC. Former CTO; comfortable inheriting unfamiliar production codebases as the senior technical owner from day one.

CERTIFICATIONS

AI Engineering Specialization - Agents, RAG, LLM applications · *ByteByteAI* · Mar 2026

Oracle Certified DevOps Professional · *Oracle* · Aug 2025

AWS Certified Cloud Practitioner · *AWS* · May 2024

Database Structures and Management with MySQL · *Coursera* · Dec 2022

Django Web Framework · *Coursera* · Dec 2022

TECHNICAL SKILLS

Frontend: JavaScript, TypeScript, React.js, Next.js, Tailwind CSS

Mobile: Flutter (iOS + Android), Codemagic CI/CD

Backend: Node.js (Express), Python (FastAPI, Django, Celery), REST APIs, GraphQL, WebSockets

Databases (NoSQL): DynamoDB (single-table and multi-table designs), MongoDB, Firebase/Firestore, Convex, Redis

Databases (SQL): PostgreSQL, MySQL, SQLite

AWS: Lambda, EventBridge, DynamoDB, EKS, ECS, S3, SQS/SNS, VPC, IAM

Other cloud: Azure, GCP (GKE, Cloud SQL, Cloud Storage), OCI

IaC & CI/CD: Terraform, Ansible, GitHub Actions, FluxCD, ArgoCD, Azure DevOps, Codemagic, Jenkins

Containers & Orchestration: Docker, Docker Compose, Kubernetes (EKS, GKE, TKG, RKE2), Helm

Messaging & Queues: AWS SQS/SNS, Kafka, RabbitMQ, Redis queues

Observability: Prometheus, Grafana, ELK Stack, Sentry

AI & Agents: LLM API integration (Claude, OpenAI), RAG systems, agent development, tool-use patterns

PROFESSIONAL EXPERIENCE

Team Lead, Core Banking Services · *Guaranty Trust Bank Ghana LTD* · Accra, Ghana

May 2025 - Present

- Built a bank-wide OTP service using Django (Python) and Next.js (React/TypeScript), integrating SMS and email gateways, to secure core banking access for 600+ internal users with structured logging to Elasticsearch via Filebeat.
- Built a national ID verification service using Django and Next.js, processing and storing ID documents in self-hosted MinIO with automated verification and customer record updates.
- Led a 5-person QA team through the bank's Basis-to-Finacle core banking migration cutover, designing and executing 10,000+ test cases across 13 core banking modules and 33 internal applications over a two-week window, covering integration, UAT, and peripheral device validation.
- Own the on-prem Finacle core banking deployment on Tanzu Kubernetes Grid, managing deployments, patching, and incident response across 800+ production pods to sustain 1.5M+ daily transactions.
- Led L2 incident response, correlating logs across internal apps and core banking microservices via the Elastic Stack for root-cause analysis and remediation.

DevOps Engineering Lead · *Guaranty Trust Bank Ghana LTD* · Accra, Ghana

Apr 2024 - Present

- Founded the bank's DevOps team of 4 engineers, replacing manual code handoffs with enforced Git workflows, containerized Kubernetes deployments, and automated release pipelines.
- Implemented CI/CD automation that cut deployment time by ~90% and increased release velocity ~5x; introduced GitOps with FluxCD for consistent, repeatable deployments; integrated Veracode and Trivy scanning for automated security analysis.
- Architected high-availability Kubernetes clusters across multiple environments running internal applications handling 3M+ daily requests, with 99.97% measured uptime inclusive of monthly scheduled maintenance.
- Managed multi-cloud infrastructure across AWS and Azure: resource provisioning, private VPC peering to on-prem systems, IAM, access control, and cost optimization.
- Automated 95% of infrastructure configuration management using Terraform, Ansible, Python, and Bash, turning multi-day manual processes into reproducible, version-controlled deployments.

Chief Technical Officer (Founding) · *MYHEALTHCOP LTD* · *Accra, Ghana*

Jan 2021 - Mar 2024

- Architected the myhealthcop.com platform end-to-end across requirements, system design, database modeling, and API development, delivering three applications: a patient-facing Flutter mobile app (iOS + Android), a Flutter app for health professionals, and a React admin web dashboard.
- Designed secure, cost-efficient AWS infrastructure running containerized microservices on EKS and ECS alongside Lambda functions for event-driven workloads, with autoscaling, load balancing, and inter-service communication.
- Built a high-throughput background processing system handling 1M+ daily jobs across payments, messaging, and notifications using SQS/SNS, with SMS, email, and payment provider integrations including multi-provider failover.
- Led performance optimizations that reduced peak request latency by 90% through Redis caching, query tuning, and indexing in PostgreSQL.
- Automated application delivery and infrastructure provisioning via CI/CD pipelines on GitHub Actions and Terraform, making deployments repeatable, auditable, and reversible.
- Led a 12-person cross-functional team (frontend, backend, mobile, product, design, QA) through full project lifecycle under Agile/Scrum.

PROJECTS

Virk Cloud (virk.cloud) · *Founder & CTO*

Aug 2025 - Present

- Architected a multi-tenant Platform-as-a-Service for startups with built-in CI/CD, fixed billing, observability, real-time log streaming, and alerting.
- Built the full management console in Next.js (React, TypeScript) with authentication, user management, real-time monitoring dashboards, and billing - inherited-and-evolved an early prototype into production.
- Manage the underlying infrastructure on virtual private servers: RKE2 cluster deployment, hardware/storage provisioning, network and traffic management, and DNS.

Next.js, React, TypeScript, Convex, Tailwind CSS, Kubernetes (RKE2), Terraform, FluxCD, Prometheus, Grafana, Python

Digital-Recon: Transaction Reconciliation Engine · *Founder*

2025 - Present

- Building a real-time transaction reconciliation engine for financial institutions, detecting discrepancies across heterogeneous data sources including relational DBs, REST APIs, Elasticsearch, and Kafka topics.
- Designed a Next.js/TypeScript triage dashboard enabling operators to define reconciliation rules and resolve mismatches with a full audit trail.
- Architected an event-driven processing layer using Kafka for upstream ingestion and Celery for scheduled and on-demand reconciliation jobs.

React, TypeScript, Next.js, Python, Celery, Kafka, Elasticsearch, PostgreSQL, Docker

Education Research and Innovation Foundry (erifoundry.org) · *Founder & Research Lead*

Oct 2025 - Present

- Leading design of AI-driven research initiatives in adaptive learning, cognitive modeling, and educational analytics.
- Coordinating cross-disciplinary research on ML and software tooling for education.
- Prototyping applications of LLMs in human-centered learning systems.

AI Agents for Infrastructure Operations · *Personal R&D*

2024 - Present

- Building specialized AI agents for QA automation, code review, and sysadmin operations, running against live Linux production servers with real workloads.
- Built AWS Lambda functions triggered by EventBridge to deliver notifications on codebase events.
- Used DynamoDB for NoSQL data modeling across personal projects, with both single-table and multi-table design patterns.

EDUCATION

BSc Computer Engineering · *Ashesi University* · *Berekuso, Ghana*

Sep 2016 - May 2020